ARTIFICIAL INTELLIGENCE

(Real-World Problem Solving Using Artificial Intelligence Approaches)

Project on:

"Recipe Retrieval"

Submitted By

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Abstract:

In this Artificial Intelligence assignment, supervised classification is used to find matches. Among the many projects which can be performed using this method, finding a recipe that uses a set of available ingredients is chosen for this individual project. The project aims to perform a text similarity algorithm to match the list of ingredients given as input to the list of ingredients in the data set.

The Real World Problem:

Nourishment in the form of food is essential for all human beings to live. One way to consume healthy is to cook food at home and making the meals vary encourages us to enjoy them more. There are a wide variety of produce and cuisines to choose from. The household members might also have various dietary restrictions and/or requirements. When one looks to cook food for oneself or their family, coming up with new and tasty recipes might not be easy. Then people end up either repeating a few recipes over and over or ordering out. To know the innumerous recipe by memory and cook up new dishes frequently isn't a realistic expectation that can be put on a home cook.

Project Aim and Objectives:

This problem can be solved using artificial intelligence. This intelligent model can be used to find recipe suggestions based on the ingredients available on hand and not needing to or minimize ingredients to be purchased. Minimizing the preparation time will also be helpful. The model can access huge datasets of recipes across countries and can be used by home cooks efficiently.

For the model to work properly, it would need to access data from multiple sources (cookbooks, videos etc.), compare them with the input values and give possible recipes within seconds.

Our objective for the project can be achieved using an Artificial model by:

1) Gather recipes from various sources like cookbooks, posts in social media websites, recipe books and so on.
2) Classify recipes according to common dietary types – vegetarian, vegan, gluten free, pescatarian, halaal, kosher etc.
3) Understand the dataset thoroughly. Clarify each ingredient used in the recipes, equipment used, time taken to prepare etc. Understand the processes, parts and timings clearly.
4) Test scenario will be giving inputs with the ingredients available and dietary requirements and receiving appropriate recipes using these as output.

Hypothesis:

The AI model will help utilize available produce to prepare varying healthy and tasty meals, thereby also reducing unnecessary food waste. Consumers can achieve this sticking to their dietary preferences and not needing to spend energy and time thinking of coming up with said new recipes. The model will encourage healthier eating habits, introduce to new cuisines, and reduce food waste (and hence save money).

Data sources reference for the project:

The dataset used for this project can be accessed through this link:

https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions

Adopted Artificial Intelligence Approach:

The model will need to traverse the data set and retrieve recipes that matches with the ingredients list given as input. We can achieve this task by using a search algorithm. We have the 2 main categories, that is, the Informed Search & the Uninformed Search. Informed Search algorithms such as the Greedy search, A* search or Graph search are widely used for pathfinding problems. They especially help weigh in the effectiveness of the generated path (or solution) by including the energy spent (or cost) on taking each particular path. For instance, in Greedy Search algorithm we expand the node closest to the goal node in the state space, where the 'closeness' is measured using a heuristic h(x).

In our case, we are not concerned about the cost of traversing the path nor where the data is stored in the system. This implies that we can use any Uninformed search algorithms such as Depth First search, Breadth First search or Uniform Cost Search.

Text similarity approach forms the basis for successful text mining operations such as searching and information retrieval (IR), text classification, information extraction (IE), document clustering, sentiment analysis, machine translation, text summarization, and natural language processing (NLP). (Didik Dwi Prasetya, 2018) The four major categories that the various text similarity approaches are divided into are string-based, corpus-based, knowledge-based and hybrid similarities.

Information Retrieval (IR) is the action where a collection of data is represented, stored and searched in order to discover knowledge as a response to a user's search. (Akram Roshdi, 2015) The main IR models can be categorized into Boolean, Vector,

Probabilistic and inference network model. In the Boolean model, a document is associated with a set of keywords and uses the Boolean algebra operators AND, OR and NOT to formulate queries. A Vector Space model attempts to rank the documents by the similarity between the query and each document. The Probabilistic Model on the other hand attempts to rank documents by their probability of relevance given a query.

Natural Language Processing techniques are used for the analysis and representation of human languages. (Cambria, 2014) NLPs can be seen as evolved from Natural Language Understanding to include the processing and understanding of semantics on top of sentics. It is a combination of common sense knowledge and common knowledge, that humans develop as they grow. In order to be able to accurately and sensibly process information, computational methods will have to be able to project semantics and sentics in time, compare them in a parallel and dynamic way with the different contexts and relate to different actors and their intentions.

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Artificial Intelligence Approach Implementation:

Our objective is to create an API that takes a list of ingredients as the input and retrieve the top 3 recipes that uses those ingredients, while minimizing additional ingredients needed. We receive the output in the format of an array of arrays.

The recipe retrieval project is a text similarity function. We compare between the list of ingredients in the input and the list of ingredients in the final recipe. In the data set, we have the recipes data file that includes all this information.

The ingredients file has a sample data as shown.

```
ingredients.head(2)
```

	raw_ingr	raw_words	processed	len_proc	replaced	count	id
0	medium heads bibb or red leaf lettuce, washed,...	13	medium heads bibb or red leaf lettuce, washed,...	73	lettuce	4507	4308
1	mixed baby lettuces and spring greens	6	mixed baby lettuces and spring green	36	lettuce	4507	4308

The recipes file has the following sample data. I

```
PP_recipes.head(2)
```

	id	i	name_tokens	ingredient_tokens	steps_tokens	techniques	calorie_level	ingredient_ids
0	424415	23	[40480, 37229, 2911, 1019, 249, 6878, 6878, 28...	[[2911, 1019, 249, 6878], [1353], [6953], [153...	[40480, 40482, 21662, 481, 6878, 500, 246, 161...	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...	0	[389, 7655, 6270, 1527, 3406]
1	146223	96900	[40480, 18376, 7056, 246, 1531, 2032, 40481]	[[17918], [25916], [2507, 6444], [8467, 1179],...	[40480, 40482, 729, 2525, 10906, 485, 43, 8393...	[1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...	0	[2683, 4969, 800, 5298, 840, 2499, 6632, 7022,...

```
recipes.head(2)
```

	name	id	minutes	contributor_id	submitted	tags	nutrition	n_steps	steps	description	ingredients	n_ingredients
83343	fettuccine and roasted vegetable pasta bake	69534	60	30716	2003-08-25	['60-minutes-or-less', 'time-to-make', 'course...	[539.5, 28.0, 23.0, 10.0, 51.0, 39.0, 22.0]	13	['preheat the oven to 180 degree c', 'grease a...	a special dish of wonderful flavours. adaptabl...	['sweet potato', 'red onion', 'zucchini', 'oli...	9
89718	garden gazpacho salad	88512	90	94272	2004-04-07	['time-to-make', 'course', 'main-ingredient', ...	[54.3, 5.0, 10.0, 1.0, 2.0, 2.0, 1.0]	3	['combine all vegetables', 'whisk together dre...	i'm ready for warm weather! this is a tasty, c...	['tomatoes', 'cucumber', 'green bell pepper', ...	14

The interactions data file is as shown below:

```
interactions.head(2)
```

	user_id	recipe_id	date	rating	review
176515	1649203	298088	2012-02-11	0	It was interesting at first bite, but after 3 ...
725061	199792	184515	2007-05-25	5	Very good even though it isn't winter. It is f...

+ Code + Markdown

We determine the size of the data files used.

```
ingredients.shape
```

```
(11659, 7)
```

```
recipes.shape
```

```
(231637, 12)
```

```
PP_recipes.shape
```

```
(178265, 8)
```

```
interactions.shape
```

```
(1132367, 5)
```

We see that the data files are rather large. Hence we choose only a portion of it for this project. Then we merge the interactions and recipes dataset using the Then we merge the interactions and recipes dataset using the recipe id. Then we merge the interactions and recipes dataset using the recipe id. It is on this new dataset that we perform the recipe retrieval.

```
interactions = interactions.sample(70000)
recipes = recipes.sample(70000)
```

+ Code + Markdown

```
data = pd.merge(interactions,recipes, right_on='id',left_on='recipe_id')
```

```
data.shape
```

```
(20735, 17)
```

Here is how the data file looks now:

```
data.head(2)
```

	user_id	recipe_id	date	rating	review	name	id	minutes	contributor_id	submitted	tags	nutrition	n_steps	steps	description	ingredients	n_in
0	510302	458619	2011-06-17	5	Very good. I made it exactly as described here...	sweet and sour carrots	458619	20	488380	2011-06-13	['30-minutes-or-less', 'time-to-make', 'course...	[139.4, 9.0, 60.0, 8.0, 2.0, 18.0, 7.0]	4	['boil sliced carrots until crips-tender', 'dr...	this is an easy and tasty new way to make carr...	['carrots', 'green onions', 'unsweetened pinea...	
1	133174	458619	2011-10-17	5	These are very flavorful and so easy to make. ...	sweet and sour carrots	458619	20	488380	2011-06-13	['30-minutes-or-less', 'time-to-make', 'course...	[139.4, 9.0, 60.0, 8.0, 2.0, 18.0, 7.0]	4	['boil sliced carrots until crips-tender', 'dr...	this is an easy and tasty new way to make carr...	['carrots', 'green onions', 'unsweetened pinea...	

We have also dropped the columns that are not necessary for our project. Information such as contributor ID, submission date is not relevant for us. A quick look at the statistics of the data is also shown.

```
data.drop(["user_id","submitted","contributor_id","id"],axis=1,inplace=True)
data.describe()
```

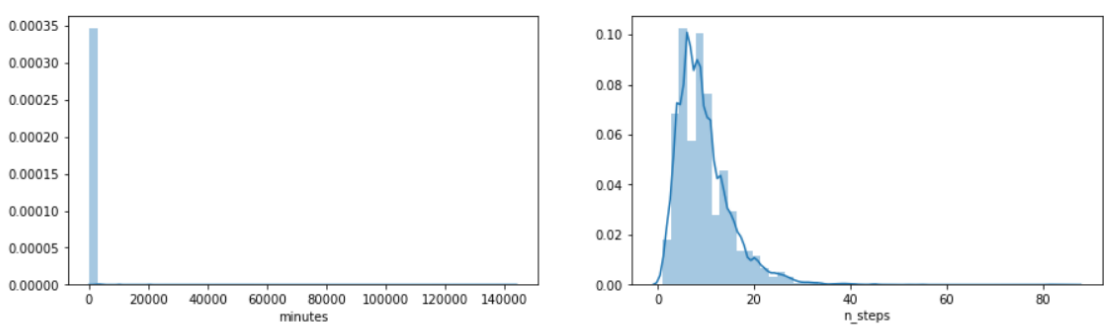
	recipe_id	rating	minutes	n_steps	n_ingredients
count	20735.000000	20735.000000	20735.000000	20735.000000	20735.00000
mean	161679.189438	4.392187	107.502532	9.617024	8.88054
std	131019.920301	1.287217	1323.209736	5.740362	3.68209
min	62.000000	0.000000	0.000000	1.000000	1.00000
25%	53720.500000	4.000000	20.000000	6.000000	6.00000
50%	124522.000000	5.000000	40.000000	8.000000	8.00000
75%	243565.000000	5.000000	70.000000	12.000000	11.00000
max	536476.000000	5.000000	144015.000000	86.000000	33.00000

+ Code + Markdown

Let's look into the dataset. We first plot the distplot for minutes taken to prepare the recipe and the number of steps in the recipe.

```
fig,ax = plt.subplots(1,2,figsize=(15,4))
sns.distplot(data["minutes"],ax=ax[0])
sns.distplot(data["n_steps"],ax=ax[1])
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x797c27b8a2d0>
```

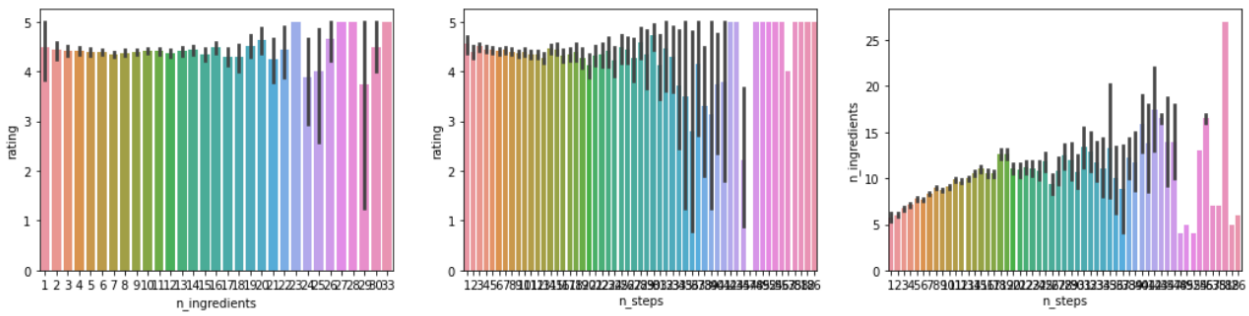


+ Code + Markdown

Next we compare the data as: Compare rating with number of ingredients, compare rating with number of steps and finally number of ingredients against number of steps.

```
fig,ax = plt.subplots(1,3,figsize=(18,4))
sns.barplot(data=data,y="rating",x="n_ingredients",ax=ax[0])
sns.barplot(data=data,y="rating",x="n_steps",ax=ax[1])
sns.barplot(data=data,y="n_ingredients",x="n_steps",ax=ax[2])
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x797c59d97c50>
```



We then check for nulls and drop them. The API for recipe retrieval is to be run on this cleaned and processed data.

Then mappers (dict) are created which is used in the API to retrieve appropriate recipes.

- r2i_map: Key = recipe id, Value = set of ingredient ids required for this recipe
- i2r_map: Key = ingredient id, Value = set of recipe ids that use this ingredient
- id2r_map: Key = recipe id, Value = recipe
- i2id_map: Key = ingredient, Value = ingredient id

And finally we create the API to retrieve the recipe which has the most similar ingredient list to the input ingredient list.

For this, we receive an output in the format of array of arrays, which has the recipe id, recipe name, the time taken for preparation, number of items used from the input list of ingredients and number of additional items needed. The recipes in output is given a score which is calculated by:

Score = numberOfIngredientsUsed(60/time taken for preparation) – numberOfAdditionalItemsNeeded(time taken for preparation/15)

The output array is sorted according to this score.

```
For Available ingredients:
['winter squash', 'mexican seasoning', 'mixed spice', 'honey', 'butter', 'olive oil', 'salt']

Recommended Recipe:
[359019, 'easiest ever peanut butter fix for one', 1, 2, 3]
With Score: 1152921504606846974.924010375

For Available ingredients:
['low sodium chicken broth', 'tomatoes', 'zucchini', 'potatoes', 'wax beans', 'green beans', 'carrots']

Recommended Recipe:
[310771, 'low carb stuffed zucchini', 2, 2, 5]
With Score: 1073741822.760644208470714532

For Available ingredients:
['spinach', 'garlic powder', 'soft breadcrumbs', 'oregano', 'onion']

Recommended Recipe:
[472668, 'country style pork ribs', 1, 2, 3]
With Score: 1152921504606846974.924010375


Overall Mean Score: 768614336762478590.869554986
```

Evaluation, Results and Discussions:

Through the process of developing the Recipe Retrieval API, we observe that the data file for recipes can be very complex. It is also often filled with information that is strictly not relevant when it concerns a search or recommendation project. Though information such as pairings or year of developing may be interesting in real life scenarios, they often prove to be a hindrance in using as a source data.

Once we filter out the unwanted components and convert the recipe into a workable data set, there are a few ways one can go about retrieving the required information. Using NLP techniques to find text similarity is the chosen method here.

Conclusion and future work

Using AI techniques for large scale searches is preferred. Relating to the project of recipe retrieval, there are many ways to further develop and scale the project. For instance, one can include different types of data for input such as images or speech. One can also work on widening the search space to include various sources such as recipe books, cooking tutorials, videos and so on. With each additional type of data in both the source material and the inputs, the ways to process the data and perform the function of matching the similar set to the recipes will use a wider variety of algorithms.

References

1. Citation: (Didik Dwi Prasetya, 2018)
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